Kairong Sun, Ph.D.

Beijing, China | skr@cau.edu.cn | kairong-sun.github.io

(ORCID) 0009-0003-7904-2536

(Google Scholar) https://scholar.google.com.hk/citations?user=fKrkHpMAAAAJ&hl

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RESEARCH PROFILE

Early-Career Researcher with experience in horticultural biology, multi-scale imaging, quantitative phenotyping, instrument-based colour measurement, and genomic/transcriptomic data integration. Applied ImageJ to quantitative biological-image analysis and an AI-assisted coding workflow to genome-annotation mapping. Interested in extending this background toward image-based and data-driven assessment of horticultural and postharvest traits.

EDUCATION

- **Ph.D. in Horticulture (Pomology)** Sep. 2022 – Jul. 2026
  China Agricultural University, Beijing, China
  - **Advisor**: Prof. Huiqin Ma
  - **Dissertation Title**: FcAGM1 Regulates Sex-Specific Floral Development in Fig (*Ficus carica* L.).
- **M.S. in Horticulture (Ornamental Horticulture)** Sep. 2019 – Jun. 2022
  Chinese Academy of Agricultural Sciences, Beijing, China
  - **Advisor**: Prof. Jingqi Xue
  - **Thesis Title**: Mechanisms Underlying DNA Demethylation-Induced Winter Flowering in Tree Peony.
- **B.S. in Horticulture (Turfgrass Science)** Sep. 2015 – Jun. 2019
  Hainan University, Haikou, China

SELECTED RESEARCH CONTRIBUTIONS

- Quantified sex-specific floral development in fig at organ, tissue, and cellular scales using developmental photography, stereomicroscopy, histology, and image-based morphometry.
- Integrated outputs from differential-expression analysis, WGCNA, phenotypic associations, and DAP-seq to prioritize FcAGM1 as a key Y-linked candidate regulator; applied AI-assisted scripting to perform downstream genomic coordinate analysis and visualization of candidate genes.
- Investigated fig peel-colour development in a first-author study, using a colour spectrophotometer and CIELAB parameters (L*, a*, and b*) to quantify colour changes and relate them to DNA-methylation and gene-expression patterns.
- Conducted controlled tree-peony forcing experiments and quantified flowering performance, including

flowering rate, flower deformity, and CIELAB petal-colour traits, contributing to a first-author publication on DNA demethylation-induced flowering.

TECHNICAL SKILLS

1. Computational Biology and AI-assisted Research

• AI-assisted computational workflows: Applied Python scripting for biological data processing, visualization, genome annotation analysis, and genomic-coordinate processing; used AI coding assistants (Claude Code/Codex) for script adaptation, debugging, and workflow optimization.

• Omics analysis and interpretation: Experienced in downstream analysis of processed omics datasets, including differential-expression analysis, GO/KEGG enrichment, WGCNA, DAP-seq peak/motif analysis, and phenotype-associated data interpretation.

• Bioinformatics and statistics: Familiar with TBtools, MEGA, iTOL, PlantCARE, and IBM SPSS Statistics; applied Student's t-tests, one-way ANOVA, and post hoc multiple-comparison tests.

• Computing environment: Basic experience with Linux command-line operations, SSH-based remote environments, and VPS-based computational workflows.

2. Imaging and Quantitative Phenotyping

• Developmental and morphological imaging using stereomicroscopy, confocal laser scanning microscopy, paraffin histology, fluorescence staining, and DAPI imaging.

• Quantitative image analysis using ImageJ/Fiji for morphometric traits, cellular measurements, stomatal characteristics, and pollen germination assessment.

• Instrument-based colour phenotyping using CIELAB parameters (L*, a*, b*) and quantitative evaluation of fruit and floral traits.

3. Molecular and Functional Validation

• Molecular cloning, vector construction, PCR/qRT-PCR, transient expression in Ficus fruit and Nicotiana benthamiana, and subcellular localization analysis.

• Functional validation using Y1H, Y2H, EMSA, dual-luciferase reporter assays, luciferase complementation imaging, and Western blotting.

• DNA methylation analysis using McrBC-PCR and bisulfite-based assays; AI-assisted protein structure prediction and molecular visualization using AlphaFold 3**.**

4. Horticultural Systems

• Fig and grape orchard management; controlled-environment forcing of tree peony and herbaceous peony; plant propagation, tissue culture, and germplasm evaluation.

• Artificial pollination, pollen viability testing and cryopreservation, fig-pollinating-wasp rearing, hybridization trials, and field phenotyping.

PUBLICATIONS

First-Author Papers

Sun, K., Wang, X., Huang, H., Wang, Y., Fan, Z., Xia, Y., Ma, H., & Song, M. (2026). FcMET1 mediates low DNA methylation and promotes peel coloring in *Ficus carica*. ***Horticultural Plant Journa****l*, 12(2), 345–355. https://doi.org/10.1016/j.hpj.2024.04.002.

Sun, K., Xue, Y., Prijic, Z., Wang, S., Markovic, T., Tian, C., Wang, Y., Xue, J., & Zhang, X. (2022). DNA demethylation induces tree peony flowering with a low deformity rate compared to gibberellin by inducing *PsFT* expression under forcing culture conditions. ***International Journal of Molecular Sciences****, 23*(12), 6632. https://doi.org/10.3390/ijms23126632.

Co-Authored Papers

Wang, X., **Sun, K**., Liao, X., Zhang, Y., Ban, Y., Zhang, X., & Song, Z. (2024). Physicochemical, antibacterial and aromatic qualities of herbaceous peony (*Paeonia lactiflora* Pall.) tea with different varieties. ***RSC Advances***, 14(20), 14303–14310. https://doi.org/10.1039/D3RA08144C.

Fan, Z.†, Wang, Y.†, Zhai, Y., Gu, X., **Sun, K.**, Zhao, D., Wang, J., Sun, P., Huang, H., He, J., Wang, Y., Flaishman, M. A., & Ma, H. (2025). ERF100 regulated by ERF28 and NOR controls pectate lyase 7, modulating fig (*Ficus carica* L.) fruit softening. ***Plant Biotechnology Journal****, 23*(7), 2611–2626. https://doi.org/10.1111/pbi.70085.

Gu, X., He, J., He, H., Wang, Y., Fan, Z., Zhao, D., **Sun, K.**, Zheng, C., & Ma, H. (2026). Efficient regeneration and genetic transformation of fig (*Ficus carica*) from stem thin cell layer explants. ***Horticultural Plant Journal****, 12*(6), 1361–1370. https://doi.org/10.1016/j.hpj.2024.11.006.

Manuscript in Preparation

Sun, K., et al. Molecular regulatory mechanism of FcAGM1 in determining unisexual flower development and style elongation in fig.

PATENTS

Huiqin Ma, **Kairong Sun**, Boshu Ouyang, Yuan Wang, Xiaojiao Gu, Hantang Huang, Yining Wang. (2026). *A Method for Obtaining Fig Seeds through Artificial Pollination*. Chinese Invention Patent, Application No.: 202610223408.8.

HONORS, AWARDS & ACADEMIC OUTREACH

- **National Scholarship for Doctoral Students** 2024.
- **Merit Student Award,** China Agricultural University 2024.
- **Australian Treasury Wine Estates Penfolds Excellence Scholarship** 2025.
- **Hong kong Jiaduobao Rural Revitalization Talent Scholarship** 2026.

SELECTED MEDIA COVERAGE

Media Outreach: Featured in an exclusive interview by ***Food & Wine Magazine (China Edition)*** regarding research on fig sex differentiation, pollinating wasp ecology, and hybrid breeding 2025

RESEARCH PROJECTS & ACADEMIC EXPERIENCES

Caprifig Germplasm Optimization & Artificial Pollination System 2022 – 2026

Conducted caprifig germplasm evaluation, propagation, and field trials; optimized year-round rearing of fig pollinating wasps (*Blastophaga psenes*) and artificial pollination procedures for fig breeding, with experience in hybridization experiments and international research collaboration.

SELECTED COURSEWORK AND TRAINING

• Modelling and Applications of Agricultural Big Data with Python — 92.5/100, China Agricultural University

• Geographic Information Systems — 94/100, Chinese Academy of Agricultural Sciences

• Spatial Analysis and Modelling with ArcGIS ModelBuilder — 90/100, Chinese Academy of Agricultural Sciences

• Multimedia and Virtual Agricultural Technology — 96/100, Chinese Academy of Agricultural Sciences

• National Computer Rank Examination (China), Level 3: Network Technology, 2017.

LANGUAGES

Chinese (native); English (working proficiency)

SELECTED INTERNATIONAL EXPERIENCE

- Portugal | 2026 —Observed local horticultural practices during visits to Lisbon and Madeira.

- Turkey | 2025 — Visited Bursa Uludağ University and discussed fig research and germplasm evaluation.

- Spain | 2024 — Observed Mediterranean fig cultivars and orchard management practices.

- Cambodia | 2023 — Observed wild Ficus species and fig–wasp interactions in tropical habitats.

REFERENCES

1. **Prof. Huiqin Ma** (Ph.D. Advisor)
   China Agricultural University
   Email: hqma@cau.edu.cn

2. **Prof. Jingqi Xue** (M.S. Advisor)
   Chinese Academy of Agricultural Sciences
   Email: xuejingqi@caas.cn.